MA 374: Financial Engineering Lab
Lab 03 Report

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Problem 1: American Option Pricing Using the Binomial Model

Methodology

American call and put options are priced using the Binomial Asset Pricing Model. The time interval $[0, T]$ is divided into M equal steps of size $\Delta t = T/M$, producing a recombining binomial tree for the stock price.

The parameters used are:

$$S(0) = 100, \quad K = 100, \quad T = 1, \quad M = 100, \quad r = 8\%, \quad \sigma = 20\%.$$

As specified in the assignment, the up and down factors are:

$$u = e^{\sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t}, \quad d = e^{-\sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t},$$

with the risk-neutral probability

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

Option prices are computed by backward induction. At each node, the American option value is the maximum of the intrinsic value and the discounted continuation value, allowing for early exercise.

Results and Sensitivity Analysis

(A) Variation with Initial Stock Price $S(0)$

As the initial stock price increases, the value of the American call option increases since the option becomes more in-the-money. Conversely, the value of the American put option decreases, approaching zero for large stock prices.

S_0.png

Figure 1: Option Prices vs Initial Stock Price $S(0)$

(B) Variation with Strike Price K

An increase in the strike price decreases the call option value, as exercising becomes less favorable. In contrast, the put option value increases because the payoff upon exercise becomes larger.

K.png

Figure 2: Option Prices vs Strike Price K

(C) Variation with Risk-free Rate r

An increase in the risk-free interest rate raises call option prices due to higher expected growth of the stock under the risk-neutral measure. Put option prices decrease as future payoffs are discounted more heavily.

r.png

Figure 3: Option Prices vs Risk-free Rate r

(D) Variation with Volatility σ

Higher volatility increases the probability of extreme stock price movements. As a result, both American call and put option prices increase with volatility.

`sigma.png`

Figure 4: Option Prices vs Volatility σ

(E) Variation with Number of Steps M

As the number of time steps increases, the binomial model provides a more refined approximation of the continuous-time model. The observed oscillations in prices diminish, and the option prices converge.

call_M.png

Figure 5: Call Option Prices vs Number of Steps M

put_M.png

Figure 6: Put Option Prices vs Number of Steps M

Problem 2: Pricing at All Nodes and Optimal Exercise Strategy

Methodology

For this problem, the American put option is analyzed using a binomial tree with $M = 5$ time steps while keeping all other parameters unchanged. The option price is computed at every node using backward induction. At each node, the intrinsic value is compared with the continuation value to determine whether early exercise is optimal.

Put Option Prices at Each Node ($M = 5$)

The computed American put option prices at each time step are:

- **$t = 0$:** [5.33]
- **$t = 1$:** [1.97, 8.86]
- **$t = 2$:** [0.35, 3.66, 14.35]
- **$t = 3$:** [0.00, 0.70, 6.74, 20.73]
- **$t = 4$:** [0.00, 0.00, 1.43, 12.27, 26.64]
- **$t = 5$:** [0.00, 0.00, 0.00, 2.90, 18.81, 32.11]

The zero values correspond to nodes where the put option is out-of-the-money and both the intrinsic and continuation values are negligible.

Optimal Exercise Strategy

At each node, early exercise is optimal if the intrinsic value exceeds the discounted continuation value. The resulting optimal exercise strategy is:

- **$t = 0$:** [False]
- **$t = 1$:** [False, False]
- **$t = 2$:** [False, False, True]
- **$t = 3$:** [True, False, False, True]
- **$t = 4$:** [True, True, False, True, True]

This strategy shows that early exercise is optimal only in sufficiently low stock price states. As maturity approaches, the exercise region expands, which is consistent with the theoretical behavior of American put options.